

CLASS XII · LECTURE 5

Functions

Option I · Two periods · Section C: characteristics of a function, with an example

By the end of this lecture you can

- Name the parts of a function: return type, name, parameters and body.
- Write a function that returns the greater of two integers.
- Distinguish the definition, the call and a prototype.
- Say what return does.

Why a function exists

A function is a named block of statements that does one job and can be used again. `main` is a function. `printf` is a function someone else wrote. A user-defined function is one you write, such as `greater`. Using functions keeps `main` readable, lets you test one job at a time, and avoids copying the same loop into five places.

DEFINITION The four parts

A function has a return type (what it sends back, or void), a name, a parameter list in brackets (the values it receives), and a body in braces (the statements that do the work). Together, the first three are the heading. The body is the definition of what it does.

```
#include <stdio.h>

int greater(int x, int y) {
    if (x > y)
        return x;
    else
        return y;
}

int main(void) {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    printf("Greater = %d\n", greater(a, b));
    return 0;
}
```

Program 5.1 `greater` returns the larger of two integers. `main` only reads, calls and prints.

Walk through 8 and 3. `main` reads them into `a` and `b`. The call `greater(a, b)` copies 8 into `x` and 3 into `y`. These are different variables. Changing `x` would not change `a`, because C passes arguments **by value**. The `return` sends 8 back, and `printf` prints it. If the numbers are equal, the `else` path returns `y`, which is correct because neither is larger.

Prototype, definition and call

Piece	What it is	Where it sits
Prototype	A declaration of the heading, ending in a semicolon, with no body	Above <code>main</code> , if the function is written below <code>main</code>
Definition	The heading plus the body	Program 5.1 defines <code>greater</code> above <code>main</code> , so no separate prototype is required
Call	The use of the function, passing actual values	<code>greater(a, b)</code> inside <code>main</code>

`return` does two things: it sends a value back to the caller, and it ends the function immediately. A `void` function may use `return`; with no value just to stop early. The last statement of a function that promised an `int` should be a `return` of an `int`.

ON THE BOARD Characteristics, for the long question

A function has a name, a return type, parameters and a body. It is called by name. It can return one value. It can be called many times. Arguments are passed by value unless you later learn pointers. It makes the program modular. Then write Program 5.1 and trace 8 and 3.

The person who writes programs in a language such as C is a **programmer**. The tool that turns the C source into machine code is a **compiler**, not an assembler and not an editor. The editor only types the text. Those two one-mark answers sit next to this lecture because this is where you meet the finished program.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. A C program is translated into machine code by a:

- A) Compiler
- B) Scanner
- C) Browser
- D) Hub

2. Comments in C are written between:

- A) `/*` and `*/`
- B) `//` only, with no other form
- C) `<` and `>`
- D) `#` and `;`

Short answers

S1. Write a function that prints the greater of two inputted numbers.

Detailed answer

L1. What are the main characteristics of a function? Describe them with an example.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. A. The editor types the source. The compiler translates it.

MCQ 2. A. `/*` comment `*/` is the form every C compiler accepts. `//` is also allowed in modern C, but the paper's symbols are `/*` and `*/`.

S1. Define `int greater(int x, int y)` which returns `x` if `x > y` and otherwise returns `y`. In main, read `a` and `b` with `scanf`, then print the value returned by `greater(a, b)`. Arguments are passed by value.

L1. A function is a named block that performs one job. Its characteristics are: it has a return type, a name and a parameter list; it has a body; it is executed only when it is called; it returns at most one value with return; and the same function can be called many times, which keeps the program shorter and easier to test. In C, the values sent in are copied (pass by value). Example: `int greater(int x, int y)` returns `x` when `x` is larger and `y` otherwise. `main` reads two integers and prints `greater(a, b)`. For 8 and 3 the returned value is 8. `printf` and `scanf` are library functions; `greater` is user-defined.